Resume

1. Biography

Dr. Jialun Liu is an associate professor in National in National Engineering Research Center for Water Transport Safety (WTS), Wuhan University of Technology. From 2006 to 2010, he studied at Wuhan University of Technology, Wuhan, China, specializing in Navigation Technology. From 2008 to 2010, he minored in Japanese at Huazhong University of Science and Technology, Wuhan, China. He obtained his BSc and BA degrees in 2010. After that, he was then recommended to continue master education at Wuhan University of Technology in Traffic Engineering and Control. In 2012, he was founded by China Scholarship Council and started his PhD research at Delft University of Technology, Delft, the Netherlands. In 2013, he received his MSc degree. From 2012 to 2017, Jialun Liu worked at the section of Ship Design, Operation, and Production in the department of Marine and Transportation Technology. From March 2017, he started working at Wuhan University of Technology. His research interests include ship maneuverability, motion control of autonomous ships, and CFD applications in maritime engineering. He received "Chutian Excellent Young Scholar" in 2017. He has published about 20 academic papers, including over 5 in international journals and 10 in international conferences. He has been appointed as reviewers of《Ocean Engineering》、《Ships and Offshore Structures》、《Journal of Marine Science and Technology》, 32nd Symposium on Naval Hydrodynamics (SNH) and other international journals and conferences.

2. Research interests

(1) Ship motion control;
(2) Mathematical modeling of ship maneuverability;
(3) Ship maneuvering performance of inland vessels;
(4) Interactions among propellers and rudders;
(5) Applications of CFD to marine engineering.

3. Publications:

3.1 International journal publications

[1] **Jialun Liu***, Robert Hekkenberg, Frans Quadvlieg, Hans Hopman. An integrated empirical manoeuvring model for inland vessels [J]. Ocean Engineering， 2017: 137，287-308. **(SCI, IF: 1.488)**

[2] **Jialun Liu***, Robert Hekkenberg, Erik Rotteveel. Hydrodynamic characteristics of multiple-rudder configurations [J]. Ships and Offshore Structures, 2017:12, 818-836. **(SCI, IF: 0.790).**

[3] **Jialun Liu***, Robert Hekkenberg. Sixty years of research on ship rudders: Effects of design choices on rudder performance [J]. Ships and Offshore Structures, 2017：12，495-512. **(SCI, IF: 0.790)**.

[4] **Jialun Liu***, Frans Quadvlieg, Robert Hekkenberg. Impacts of rudder profiles on manoeuvring performance of ships [J]. Ocean Engineering, 2016:124, 226-240. **(SCI, IF: 1.488)**.

[5] **Jialun Liu***, Robert Hekkenberg, Erik Rotteveel, Hans Hopman. Literature review on evaluation and prediction methods of inland vessel manoeuvrability [J]. Ocean Engineering, 2015: 106, 458－471. **(SCI, IF: 1.488)**.

3.2 Domestic journal publications

[1] 赵丙乾，方昭昭，肖曙明，**刘佳仑**，基于 FINE/Marine 的螺旋桨水动力性能研究 [J], 船舶工程, 2018:4,45-50

[2] 刘明俊, **刘佳仑**, 周立等. 基于排队论的三峡库区锚地容量分析 [J]. 武汉理工大学学报 (交通科学与工程版), 2013, 37(1):35-38, 43.

[3] 刘明俊, **刘佳仑**, 周立等. 深水航道上延方法及效益分析 [J]. 武汉理工大学学报(交通科学与工程版), 2012, 36(6):1232-1235.

[4] 周俊安, **刘佳仑**, 刘明俊等. 长江干线航道信息发布模式与发布手段研究 [J]. 武汉理工大学学报(交通科学与工程版), 2012, 36(5):1017-1020.

[5] 邓良爱, 刘明俊, **刘佳仑**等. 对长江干线航道信息指标系统的思考 [J]. 武汉理工大学学报(交通科学与工程版), 2012, 36(4):773-777.

[6] 刘明俊, **刘佳仑**, 周立等. 取、排水工程船撞风险源识别与防护措施探讨 [J]. 武汉理工大学学报(交通科学与工程版), 2012, 36(3):599-602.

[7] 刘明俊, 徐言民, **刘佳仑**. 桥梁防船撞预控技术研究综述 [J]. 桥梁防船撞研究与工程应用. 人民交通出版社, 2011.

[8] 刘明俊, 潘俊, **刘佳仑**等. 长江干线航道信息发布平台的构建设想 [J]. 交通信息与安全, 2010, 28(6):85-87, 90.

3.3 Monographs

[1] **Jialun Liu**, Impacts of Rudder Configurations on Inland Vessel Manoeuvrability，Delft University of Technology, ISBN 978-94-6233-5622, pp. 1-198.，2017.2.27

[2] Robert Hekkenberg，**Jialun Liu***，Developments in inland waterway vessels，Inland Waterway Transport: Challenges and Prospects, Routledge, ISBN 978-1-138-82671-7, pp. 142-167, 2016.7.1

3.4 International conferences

[1] Chen Chen, Feng Ma, **Jialun Liu**, Xinping Yan, Xianqiao Chen, A novel path planning approach for unmanned ships based on deep reinforcement learning [C], Conference: Conference on Data Science and Knowledge Engineering for Sensing Decision Support (FLINS 2018), 2018.8.21-24, Belfast, Northern Ireland, United Kingdom.

[2] Feng Ma, Bolan Zhang, Chen Chen, **Jialun Liu***, Design and Application of Multi-function Marine Simulators for the Future [C], International Conference on Marine Simulation and Ship Maneuverability (MARSIM 2018), 2018.8, Halifax, Canada.

[3] Le Wang, Qing Wu, Shijie Li, **Jialun Liu***, Ship Course Control based on BAS Self-Optimizing PID Algorithm [C], International Conference on Marine Simulation and Ship Maneuverability (MARSIM 2018), 2018.8, Halifax, Canada.

[4] **Jialun Liu***, Yuanyuan Wang, Feng Ma, Shijie Li, Impacts of Rudders on the Performance of a Benchmark Inland Vessel in the Rhine [C], International Conference on Marine Simulation and Ship Maneuverability (MARSIM 2018), 2018.8, Halifax, Canada.

[5] **Jialun Liu***, Robert Hekkenberg, Bingqian Zhao, RANS Study on Hydrodynamic Characteristics of Flapped Rudders [C], 37th International Conference on Ocean, Offshore & Arctic Engineering (OMAE 2018), 2018.6.17-2018.6.22, Madrid, Spain.

[6] Shijie Li, Rudy R. Negenborn, **Jialun Liu***, Stimulating inland waterway transport between seaports and the hinterland from a coordination perspective, Transportation Research Board 97th Annual Meeting (TRB 2018), 2018.1.7-2018.1.11, Washinton DC, USA.

[7] **Jialun Liu***, Robert Hekkenberg, Zhonglian Jiang, Xiumin Chu, Mesh properties for RANS simulations of aerofoil-shape rudder hydrodynamics, 10th International Workshop on Ship and Marine Hydrodynamics (IWSH 2017), 2017.11, Keelung, Taiwan, P. R. China.

[8] **Jialun Liu***, Robert Hekkenberg, Interaction effects on hydrodynamic characteristics of twin rudders [C]. International Conference on Maritime Technology, 2016: 1-7, Harbin, P. R. China.

[9] **Jialun Liu***, Robert Hekkenberg, 3D RANS simulations of shallow water effects on rudder hydrodynamic characteristics [C]. International Conference on Maritime Technology, 2016: 33-38, Harbin, P. R. China.

[10] **Jialun Liu***, Frans Quadvlieg, Robert Hekkenberg. Impacts of rudder profiles on ship manoeuvrability [C], International Conference on Marine Simulation and Ship Manoeuvrability (MARSIM'15), 2015, New Castle, UK.

[11] **Jialun Liu***, Robert Hekkenberg. Hydrodynamic characteristics of twin-rudders at small attack angles, 12th International Marine Design Conference (IMDC), 2015: 177-188, Tokyo, Japen.

[12] **Jialun Liu***, Robert Hekkenberg, Erik Rotteveel. A proposal for standard manoeuvres and parameters for the evaluation of inland vessel manoeuvrability [C]. European Inland Waterway Navigation Conference (EIWN), 2014, Budapest, Hungary.

[13] Erik Rotteveel, Robert Hekkenberg, **Jialun Liu***. Design guidelines and empirical evaluation tools for inland vessels [C]. European Inland Waterway Navigation Conference (EIWN), 2014, Budapest, Hungary.

[14] **Jialun Liu***, Mingjun Liu, Yanmin Xu. Flow field simulation of bridge group area considering cross [C]. International Conference on Marine Simulation and Ship Manoeuvrability (MARSIM'12), 2012.

3.5 Software copyrights

[1] 刘佳仑, 夏鑫, 内河船舶操纵性仿真软件 V1.0, 2018SR665906, 2018.